PROGRAM 9: Develop a Simple AI Assistant Using External Documents

AIM

To develop an AI assistant that answers user questions by retrieving relevant information from external text documents.

ALGORITHM

1. Install the required libraries.
2. Load the embedding and text-generation models.
3. Load text documents from external files.
4. Convert document text into embeddings.
5. Store the embeddings in a vector database.
6. Accept a user question.
7. Retrieve the most relevant document passages.
8. Generate an answer based on the retrieved information.
9. Display the answer.

CODE

Cell 1: Install libraries

```
!pip -q install sentence-transformers faiss-cpu transformers sentencepiece
```

Cell 2: Python program

```
import os
import faiss
import numpy as np
import torch
from sentence_transformers import SentenceTransformer
from transformers import pipeline


# Create example external text files
os.makedirs("documents", exist_ok=True)


with open("documents/ai.txt", "w") as f:
    f.write(
        "Artificial intelligence enables machines to perform "
```

```
        "tasks requiring human intelligence."
    )

with open("documents/ml.txt", "w") as f:
    f.write(
        "Machine learning is a branch of artificial intelligence "
        "that learns patterns from data."
    )

with open("documents/nlp.txt", "w") as f:
    f.write(
        "Natural language processing helps computers understand "
        "and process human language."
    )

# Load documents from files
texts = []
filenames = []

for filename in os.listdir("documents"):
    if filename.endswith(".txt"):
        with open(os.path.join("documents", filename), "r") as f:
            texts.append(f.read())
            filenames.append(filename)

# Load models
embedder = SentenceTransformer("all-MiniLM-L6-v2")

generator = pipeline(
    "text2text-generation",
```

```
    model="google/flan-t5-small",
    device=0 if torch.cuda.is_available() else -1
)

# Create vector database
vectors = embedder.encode(
    texts,
    convert_to_numpy=True
).astype("float32")

faiss.normalize_L2(vectors)

index = faiss.IndexFlatIP(vectors.shape[1])
index.add(vectors)

# Ask assistant
question = "What is machine learning?"

q_vector = embedder.encode(
    [question],
    convert_to_numpy=True
).astype("float32")

faiss.normalize_L2(q_vector)

scores, ids = index.search(q_vector, 2)

context = "\n".join(texts[i] for i in ids[0])

prompt = (
```

```
    f"Use this document context to answer the question.\n"
    f"Context: {context}\n"
    f"Question: {question}\n"
    "Answer:"
)

answer = generator(
    prompt,
    max_new_tokens=100,
    do_sample=False
)[0]["generated_text"]

print("AI Assistant")
print("Question:", question)
print("\nRetrieved Documents:")

for i in ids[0]:
    print(f"{filenames[i]}: {texts[i]}")

print("\nAnswer:", answer)
```

INPUT

What is machine learning?

OUTPUT

AI Assistant

Question: What is machine learning?

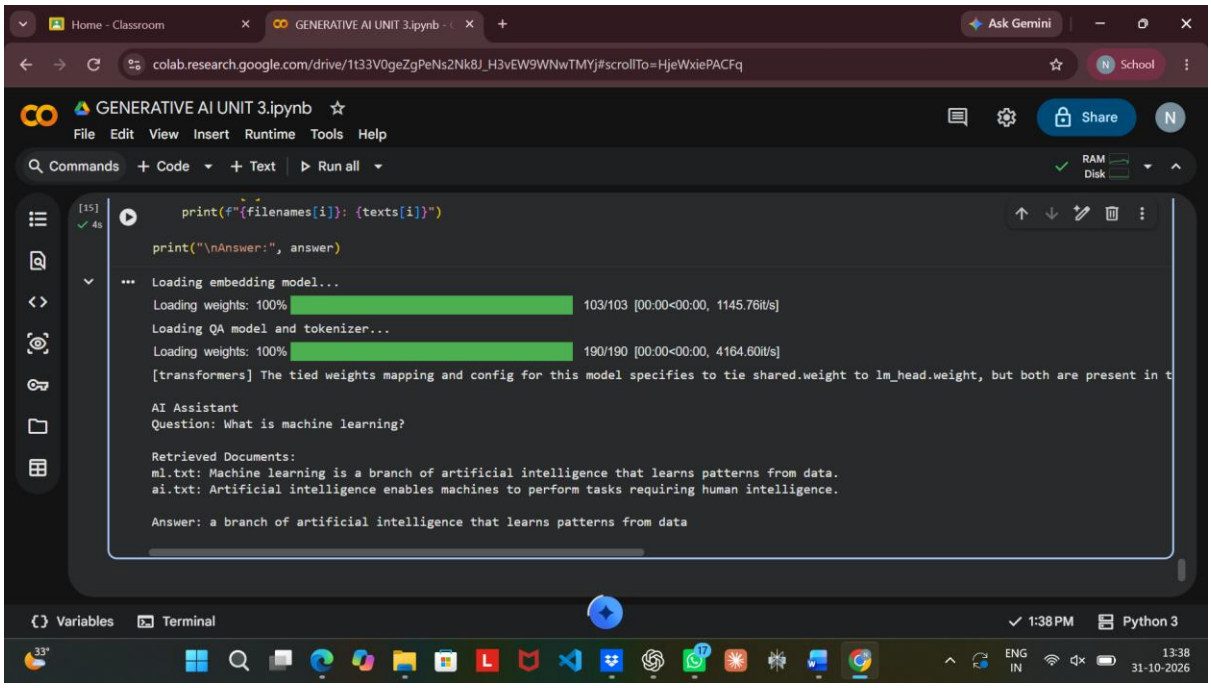
Retrieved Documents:

[Relevant external text files]

Answer:

Machine learning is a branch of artificial intelligence

that learns patterns from data.



RESULT

A simple AI assistant was developed to answer questions using information retrieved from external text files.